

Image Formation

Technical Notes

Rodrigo Kang

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1 Computer Vision as an Inverse Problem

Many problems in computer vision can be understood as **inverse problems**. An image is the result of a physical process in which a three-dimensional scene is projected onto a two-dimensional image plane and converted into digital measurements. Computer vision attempts to reason in the opposite direction: given these measurements, we want to recover properties of the scene that produced them.

Schematically, the image formation process can be represented as

$$S \xrightarrow{F} I,$$

where S represents the properties of the scene, F represents the image formation process, and I is the resulting image. The scene properties may include quantities such as geometry, depth, object shape, surface orientation, reflectance, illumination, and motion.

The corresponding inverse problem is

$$I \xrightarrow{\text{inference}} S.$$

The difficulty is that information is generally lost or mixed during image formation. As a consequence, different physical scenes may be consistent with the same image measurements. The inverse problem is therefore often **ill-posed**.

A problem is said to be well-posed, in the classical sense of Hadamard, when:

1. a solution exists;
2. the solution is unique; and
3. the solution depends continuously on the observed data.

If one or more of these conditions fails, the problem is said to be **ill-posed**.

In computer vision, this ambiguity can arise in several ways. Geometrically, projecting a three-dimensional scene onto a two-dimensional image can discard information about the spatial structure of the scene, such as depth. Photometrically, the measured intensity at a pixel can depend simultaneously on illumination, surface reflectance, camera optics, and sensor characteristics. Different configurations of these quantities may therefore produce the same, or very similar, image measurements.

2 Geometric Image Formation

The first question is how the geometry of a three-dimensional scene is represented in a two-dimensional image. Consider two objects of the same size placed at different distances from a camera. Although their physical dimensions are identical, the more distant object appears smaller in the image.

This effect can be explained using the **pinhole camera model**, a simple geometric model of image formation in which light rays from the scene pass through a single point, called the **camera centre**, before reaching the image plane.

2.1 Pinhole Camera Model

Consider a point in three-dimensional space,

$$P = (X, Y, Z),$$

expressed relative to the camera centre. The Z -axis points in the viewing direction of the camera, while X and Y describe the horizontal and vertical position of the point.

In a physical pinhole camera, the image plane lies behind the camera centre, producing an inverted image. In computer vision, it is convenient to use an equivalent **virtual image plane** located in front of the camera centre. This convention avoids the sign reversal while preserving the geometry of perspective projection.

Let the virtual image plane be located at a distance f from the camera centre, where f is the **focal length**, and let

$$p = (x, y)$$

denote the projection of P onto this plane.

The relationship between the 3D point and its projection follows directly from similar triangles. In the horizontal direction,

$$\frac{x}{f} = \frac{X}{Z},$$

which gives

$$x = f \frac{X}{Z}. \quad (1)$$

The same argument in the vertical direction gives

$$y = f \frac{Y}{Z}. \quad (2)$$

Therefore, the projection of $P = (X, Y, Z)$ onto the image plane is

$$(x, y) = \left(f \frac{X}{Z}, f \frac{Y}{Z} \right). \quad (3)$$

This mapping is known as **perspective projection**.

To see its effect, consider the point

$$P = (2, 1, 10)$$

and a camera with focal length $f = 5$. From Equation 3,

$$x = 5 \frac{2}{10} = 1, \quad y = 5 \frac{1}{10} = 0.5.$$

Therefore,

$$(2, 1, 10) \longrightarrow (1, 0.5).$$

Now consider a point with the same X and Y coordinates but twice the depth,

$$P' = (2, 1, 20).$$

Its projection is

$$x' = 5 \frac{2}{20} = 0.5, \quad y' = 5 \frac{1}{20} = 0.25,$$

and therefore

$$(2, 1, 20) \longrightarrow (0.5, 0.25).$$

Doubling the depth halves both image coordinates relative to the optical axis. More generally, for fixed X, Y , and f ,

$$x, y \propto \frac{1}{Z}.$$

This is the geometric reason why objects appear smaller as their distance from the camera increases.

The effect can also be seen by projecting the same object at different depths. Figure 1 shows the projection of a square whose physical dimensions remain unchanged while its distance from the camera increases.

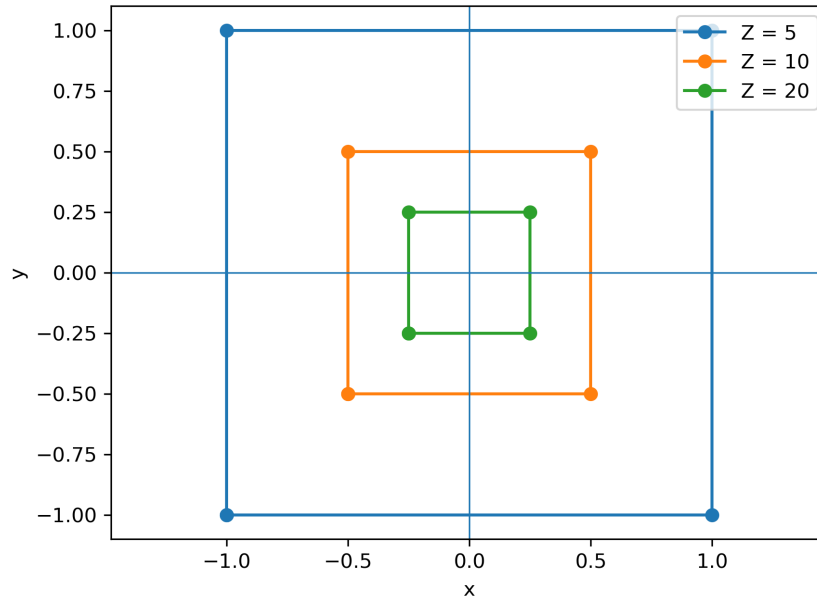


Figure 1: Perspective projection of the same square at different depths.

As the depth Z increases, the projected square contracts towards the optical axis. The change in apparent size follows directly from the $1/Z$ dependence in Equation 3.

The focal length has a different effect. For a fixed 3D point,

$$x, y \propto f.$$

Increasing f therefore increases the size of the projection on the image plane, while decreasing f reduces it.